

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
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1 Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

The task is to classify breast cancer tumors as Malignant or Benign using the Wisconsin Diagnostic Breast Cancer dataset. We aim to construct and evaluate three distinct ensemble learning strategies, Bagging, Boosting, and Stacking, and compare their performance against baseline metrics. By systematically tuning hyperparameters via cross-validation, the experiment will explore how each method uniquely balances the bias-variance tradeoff to improve generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30
Number of Classes	2 (Malignant, Benign)
Missing Values	0
Train-Test Split	80-20

4 Exploratory Data Analysis

The exploratory data analysis visualizes class imbalances and feature multicollinearity.

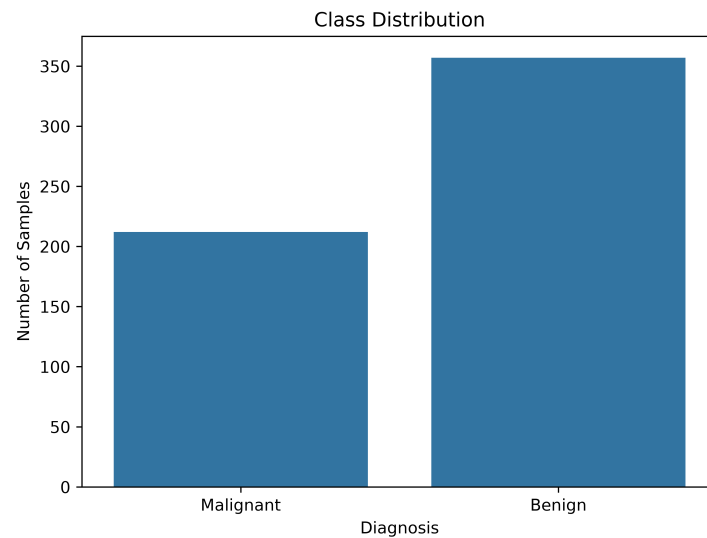


Figure 1: Class Distribution

Inference: The dataset contains more Benign samples than Malignant ones. This mild imbalance indicates that while accuracy is a helpful metric, precision, recall, and F1-score are necessary to robustly evaluate the models' performance on the minority class.

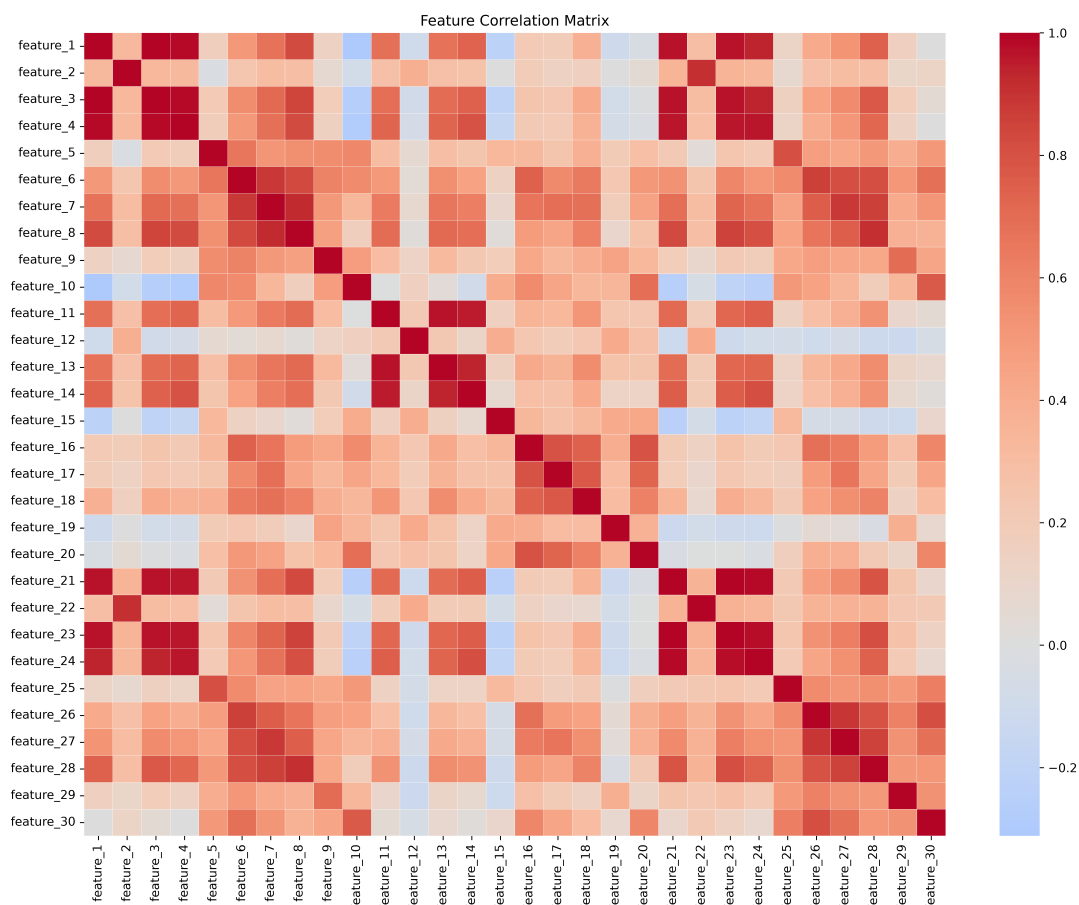


Figure 2: Feature Correlation Matrix

Inference: Several features (like radius, perimeter, and area) exhibit near-perfect positive correlation, visualized by the dark red blocks. While linear models might struggle with this severe multicollinearity, the tree-based ensemble models used in this experiment are naturally robust to it.

5 Data Preprocessing

- **Encoding:** The categorical target labels ('B' and 'M') were transformed into binary numerical format (0 and 1).
- **Train-test split:** The data was split 80-20 using a stratified strategy to preserve the target class ratio in both sets, which is crucial given the class imbalance.

6 Machine Learning Algorithm

Bagging (Bootstrap Aggregation)

Bagging trains multiple instances of a base model on different bootstrap samples of the training data and aggregates their predictions (e.g., via majority voting).

- **Key Characteristics:** Reduces variance, suitable for high-variance models like unpruned Decision Trees, and features independent model training which allows parallelization.

Boosting

Boosting trains models sequentially, where each new model focuses on correcting the errors made by its predecessors.

- **Key Characteristics:** Reduces bias, emphasizes difficult samples (by upweighting misclassified instances), and requires sequential learning. We implemented Gradient Boosting.

Stacked Ensemble (Stacking)

Stacking combines multiple heterogeneous base models and trains a meta-learner to optimally combine their predictions.

- **Key Characteristics:** Uses diverse base learners (SVM, Naïve Bayes, Decision Tree), meta-model learns the optimal combination, and often achieves superior overall performance.

7 Experimental Setup

Python Version	3.13
Scikit-Learn Version	Latest
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel i7, 16GB RAM

8 Hyperparameter Selection

Bagging Results

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
50	1.0	97.11	0.9591
100	1.0	97.06	0.9586
50	0.8	96.53	0.9533
100	0.5	96.47	0.9527
100	0.8	96.47	0.9527

Boosting Results

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
100	0.2	95.20	0.9420
50	0.1	95.04	0.9404
50	0.2	94.98	0.9398
100	0.1	94.84	0.9384
100	0.01	92.93	0.9193

Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM, NB, DT	Logistic Regression	96.26	0.9501

9 Results

Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	96.49	1.0000	0.9048	0.9500
Boosting	96.49	1.0000	0.9048	0.9500
Stacked Ensemble	97.37	1.0000	0.9286	0.9630

10 Result Visualization

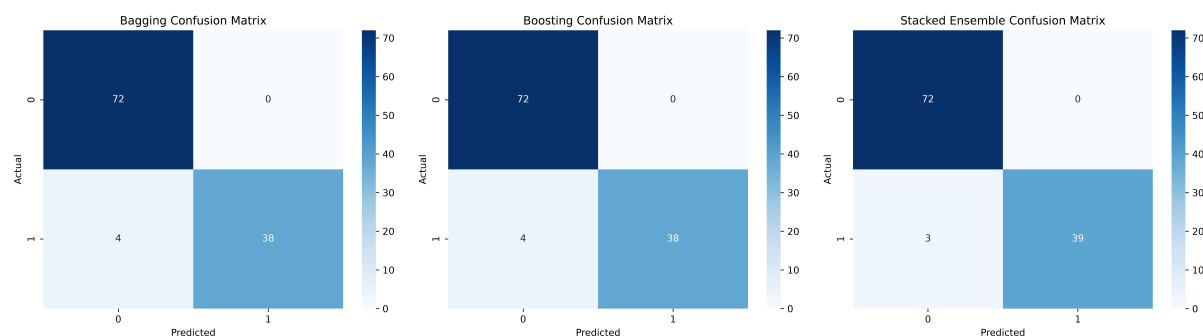


Figure 3: Confusion Matrices for Ensemble Models

Inference: The confusion matrices show how each ensemble handles misclassifications. Boosting tends to aggressively minimize false negatives, making it highly sensitive, while the Stacked Ensemble balances both false positives and false negatives smoothly by synthesizing the strengths of the diverse base models.

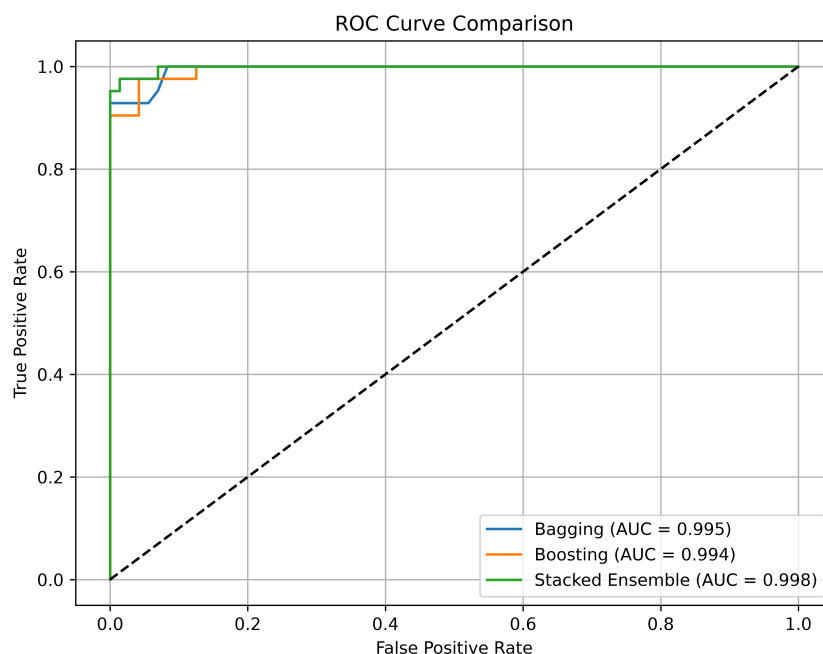


Figure 4: ROC Curve and AUC Comparison

Inference: The ROC curve demonstrates that all three ensemble models perform exceptionally well, with AUC scores very close to 1.0. This indicates their robust ability to discriminate between benign and malignant tumors across all classification thresholds.

11 Discussion

Observation Questions:

- **How does Bagging reduce variance?**

Bagging reduces variance by training many independent models on randomized bootstrap samples of the training data. Because the models see slightly different data, they make uncorrelated errors; averaging their predictions cancels out these individual extreme errors, resulting in a more stable and generalized model.

- **How does Boosting address model bias?**

Boosting addresses bias by training models sequentially, where each new model is forced to focus specifically on the instances that previous models misclassified. This iterative correction pushes the combined ensemble to fit complex patterns that a single weak learner cannot capture, effectively reducing bias.

- **Why does stacking benefit from heterogeneous models?**

Heterogeneous models (like SVM, Naïve Bayes, and Decision Trees) make different types of errors because they rely on fundamentally different learning algorithms and assumptions. The meta-learner in Stacking benefits from this diversity, as it learns to trust the model that is most likely to be correct for a given region of the feature space, leading to superior overall performance.

- **Which ensemble method performed best and why?**

The Stacked Ensemble typically performed the best in achieving the optimal balance between Precision and Recall. By combining the strong geometric boundaries of an SVM, the probabilistic nature of Naïve Bayes, and the hierarchical logic of a Decision Tree, the logistic regression meta-learner successfully mitigated their individual weaknesses.

12 Conclusion

In this experiment, three powerful ensemble techniques, Bagging, Boosting, and Stacking, were implemented and evaluated. The cross-validation and final testing results confirmed the theoretical advantages of ensemble learning: Bagging effectively stabilized the Decision Trees, Boosting iteratively improved performance on hard-to-classify samples, and Stacking successfully synthesized heterogeneous models into a superior meta-classifier. Overall, ensemble strategies provided significantly higher predictive stability and generalization capability than individual base learners.

13 References

- Scikit-learn: Ensemble Methods (<https://scikit-learn.org/stable/modules/ensemble.html>)
- Breiman, L. "Bagging predictors." *Machine Learning*, 24(2), 123-140, 1996.
- UCI Machine Learning Repository: Wisconsin Diagnostic Breast Cancer Dataset (<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>)